Milestone 01: Literature Review for Autonomous Ackermann Vehicle

Course: MCTR1002 - Autonomous Systems

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***Abstract*—This report fulfills the Milestone 01 research requirements for the MCTR1002 Autonomous Systems project. It reviews state-of-the-art implementations of autonomous ground vehicles focusing strictly on the Ackermann steering geometry. The review evaluates 2-3 key papers across four core modules: software architecture, sensor localization, navigation and planning, and vehicle control.**

I. Introduction

The development of autonomous vehicles requires a robust integration of hardware and software. For vehicles utilizing Ackermann steering kinematics, standard differential-drive approaches are insufficient. This review synthesizes literature on modern ROS-based architectures, Extended Kalman Filter (EKF) localization, Model Predictive Control (MPC), and Frenet-frame planning to guide the implementation on a Raspberry Pi / ROS 2 Jazzy stack.

II. Autonomous Car Architecture

The software architecture of an autonomous vehicle acts as the backbone connecting perception, planning, and control. Research by [1] highlights the necessity of secure, decentralized frameworks in modern connected vehicles. In the context of the Robot Operating System (ROS 2), a modular hierarchy is employed. This architecture decouples high-level computational tasks (such as global path planning and SLAM) from low-level reactive tasks (such as motor control). By isolating nodes communicating via the Data Distribution Service (DDS), the system ensures real-time reliability, which is critical when bridging the compute gap between an onboard Raspberry Pi and a base station PC.

III. Localization Module

Accurate state estimation is vital for an Ackermann vehicle, where the turning radii of the inner and outer wheels differ. The mathematical basis for this geometry, as optimized in [2], dictates the relationship between the steering angles:

$$\cot(\delta_o) - \cot(\delta_i) = \frac{T}{L}$$

where δ_o and δ_i are the outer and inner steering angles, T is the track width, and L is the wheelbase.

Relying solely on wheel encoders leads to accumulated drift. Promkaew et al. [3] propose an EKF specifically formulated for Ackermann robots to fuse encoder odometry with IMU data. The EKF predicts the vehicle state using the non-holonomic kinematic model and updates it using sensor measurements. Furthermore, Georgiev [4] demonstrates the successful deployment of the `robot_localization` package in ROS 2, proving that fusing standard IMU, Odometry, and GPS topics yields sub-centimeter positional accuracy suitable for lane-keeping applications.

IV. Navigation and Planning Module

Navigation for non-holonomic vehicles must satisfy both environmental constraints (obstacle avoidance) and kinematic constraints (minimum turning radius).

For trajectory generation, [5] introduces real-time planning utilizing the Frenet coordinate frame. By decoupling the vehicle's motion into longitudinal and lateral components relative to a reference path, the system can generate smooth, dynamically feasible trajectories. This is highly effective for lane-changing scenarios.

For dynamic environments, Model Predictive Control (MPC) serves as both a local planner and an execution module. Research in [6] validates an MPC algorithm that predicts future vehicle states over a finite horizon to execute emergency obstacle avoidance. By formulating the planning task as an optimization problem subjected to the vehicle's steering limits, the MPC guarantees collision-free navigation.

V. Control Module

The control module is bifurcated into longitudinal (speed) and lateral (steering) control. To maintain stability during high-speed cornering, [7] investigates electronic differential control systems. The authors employ a combination of PID and Backpropagation (BP) neural networks to dynamically distribute torque to the driving wheels based on the Ackermann steering angle, minimizing slip.

For lateral control (lane keeping), while Pure Pursuit is a computationally light standard, [6] demonstrates that MPC provides superior path-tracking performance. By explicitly accounting for actuator delay and kinematic constraints within its cost function, MPC minimizes the cross-track error far more effectively than traditional geometric controllers, ensuring the vehicle does not oversteer out of its lane.

VI. Conclusion

The literature confirms that developing an autonomous Ackermann vehicle requires specialized approaches. The integration of EKF for sensor fusion [3], [4], Frenet frames for smooth trajectory generation [5], and MPC for combined local planning and lateral control [6] represent the most robust methodology. These findings will serve as the architectural blueprint for the ROS 2 Jazzy implementation in the subsequent project milestones.

References

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